



EnsembleNet: a novel deep ensemble learning model with GANBERT and BiLSTM for automated fake news detection

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Abstract

Nowadays, the expansion of online social networks resulting in the spread of social news. Some social networks including Twitter, Instagram and Facebook have been impacted by user-generated fake news. Unfortunately, some individuals utilize unethical tactics to increase their links and reputation. Recent information on online social networks is often inaccurate and can mislead other users. Fake news is purposely propagated to deceive readers, making it harder to detect through shared information. This research proposes EnsembleNet: A Novel Deep Ensemble Learning Model with GANBERT and BiLSTM for Automated Fake News Detection (DEL-BiLSTM-AFND) to tackle this problem. Initially, the input data are gathered from ISOT Fake news dataset and Politi Fact dataset, followed by preprocessing for tokenization. The Generative Adversarial Networks-Bidirectional Encoder Representations from Transformers (GANBERT) is used to generate high quality word embeddings from the news articles. Then the Convolutional Neural Network is used to extract key textual features from word embeddings. Then the Bidirectional Long Short-Term Memory is considered to classify news articles fake or real depending upon the sequential dependencies. At last, the performance of proposed DEL-BiLSTM-AFND approach offers 20.35, 24.62 and 21.56% higher accuracy and 23.10, 21.19 and 24.21% higher f1- score compared with existing models: content-dependent transfer learning method for fake news detect in social media (TLA-FND), Word embedding and categorization models and their effects on fake

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news detection (WEC-FND) and Modeling Context and Content Features for Fake News Detection (MCCF-FND) methods, respectively.

Keywords Bidirectional long short-term memory · Fast guided median filter · Automated fake news detection · Convolutional neural network

1 Introduction

Since mid-1990, communication patterns have shifted. Online social networks like Facebook, WhatsApp, Instagram, and Twitter facilitate the sharing of real-time information among users across networks [1, 2]. Online social networks are becoming popular for communication and information sharing due to their effortlessness of use, rapid transformation and low cost. Currently, most social media users get their news online. However, as OSNs gain popularity, the Internet has developed as an ideal medium for communication and the transmission of bogus news [3–5]. Fake news can take various forms, including false information, reviews, rumors, ads, political speech, and satire. Fake news is spreading more on social media than in traditional media outlets. Multiple propagandists manipulated information to spread political and other influential messages over the network [6, 7]. During the 2019 Indian general election, fake accounts were created to propagate false information on Twitter and Facebook [8, 9]. Furthermore, users develop and broadcast incorrect and fantastic information on social networks. It posed a threat to numerous communities and negatively impacted people through advertising, online purchasing, and social messaging. Misinformation on social media can cause confusion and distrust among users [10, 11]. Detecting and identifying fake news on social media (SM) networks can be tough. Fake news has a significant impact on millions of users and the real world. On social media, fake news has been a problem [12, 13]. Companies and individuals utilize various SM networks to promote their products and establish a reputation. These operations encourage numerous users to share and enjoy the news [14, 15]. This approach also spreads bogus news throughout networks [16]. Fake news can vary in content, style, and media platform, and often distorts language. Furthermore, this item mocked the actual story. Fake news might contain accurate evidence to back up a false assertion. Many people use manual fact-checking tools like FaceChek.org and PolitiFact.com [17, 18]. These websites help spot bogus news on the internet. However, all of these software's require expert analysis to provide a quick response [19, 20]. Fact-checking systems typically focus on political problems. Also, users generate, share, like, and comment on a significant amount of material on social media platforms [21]. Fake persons propagate false information throughout social media platforms through many posts. The abundance of shared content on social media might make it challenging to identify. Fake news cannot go unnoticed [22, 23].

Particularly in humorous content, many classic models mainly rely on static embeddings or superficial linguistic features, which are insufficient for capturing contextual and semantic nuances [24, 25]. Word2Vec + LSTM [26] and the Tri-Algo Guardian Ensemble [24] are two examples of models that do quite well on isolated datasets but struggle with contextual understanding and lack cross-domain adaptability. Moreover, sophisticated models like AA-BiLSTM [25] and FNNNet [28] are computationally demanding and frequently overlook multimodal inputs, which make them inappropriate for real-time deployment. Multimodal integration is still a major problem because current frameworks are unable to successfully combine metadata, user behavior and textual data into a coherent learning process. There are trade-offs between specificity and precision in a number of methods; for example, models

with higher recall frequently produce large false-positive rates by misclassifying authentic content as fraudulent [28, 30]. This work aims to fill the gap by providing a real-time, scalable, and generalizable solution for fake news detection. By leveraging adversarial training GANBERT, convolutional feature learning (CNN) and sequential modeling (Bi-LSTM), the proposed approach significantly advances the state-of-the-art in both methodology and performance.

The aim of this research is to combine sequential dependency analysis, spatial feature extraction and contextual understanding to make deep ensemble mode that is both scalable and robust enough to detect false news, also increase classification accuracy, decrease false positives and guarantee adaptability across a variety of digital information by combining GAN-based embeddings with convolutional and recurrent architectures.

The key contributions of this paper include:

- Proposed a new ensemble deep learning model (DEL-BiLSTM-AFND) that combines CNN for spatial feature extraction, Bi-LSTM for sequential modeling, and GANBERT for contextual word embeddings to achieve reliable false news detection.
- Designed an effective text preprocessing pipeline that includes BERT-compatible tokenization, stemming and text cleaning to improve input data quality and model performance.
- The proposed model exceeds current approaches across the metrics on two benchmark datasets (ISOT and PolitiFact), demonstrating superior performance.
- Adversarial training of GANBERT addresses class imbalance and semantic ambiguity, improving the identification of subtle as well as minority-class fake news occurrences.

Remaining paper is structured as follows: Sect. 2 presents literature survey, Sect. 3 explains proposed technique, Sect. 4 provides outcome, Sect. 5 offers conclusion.

2 Literature survey

Among various researches for automated fake news detection across digital platforms; some of the recent research studies are reviewed here as follows:

Vineetha and Nirmala [24], presented Tri-Algo guardian ensemble method in social media for fake news identification. It uses several datasets to assess classification techniques, each of which has unique features and challenges. While different word embedding approaches capture linguistic and contextual aspects, advanced preprocessing techniques improve the quality of the data. The oversampling and class weighting techniques were used to solve class imbalance. Ensemble models provide robustness and scalability by routinely outperforming individual classifiers. It contains higher accuracy with lower precision.

Merryton and Gethsiyal Augusta [25] presented an attribute-wise attention method with BiLSTM for effectual fake news recognition. The goal was to make a novel deep architecture that could capture fundamental information regarding the content in the format of summarized feature vector representation. With the aid of convolutional and average pooling layers, AA-BiLSTM incorporates Bi-LSTM and attribute-wise attention strategy that relies on CNN to extract high-level data. Following the fusing process, the model uses the dependent characteristic of the attention process to consider the word or sentence. It offers high precision and low accuracy.

Mallik and Kumar [26] suggested word2Vec with LSTM dependent deep learning strategy to detect context-free false news. By using Word2Vec embedding with LSTM, a deep learning basis hybrid method was presented. Word2Vec embedding to obtain vector depictions of news

excerpts. It helps create feature vectors for news stories that are independent of context and data. It attained more f1 score and less accuracy.

Rustam et al. [27] developed fake news finding under enhanced features via text-to-image transformation along customized modes. It focused on identifying fake news utilizing AI-dependent method to get higher accuracy under the notion of text conversion as imageries. It converts the text as a common picture format, enhancing the feature space by improving machine learning models' performance. Comprehensive tests incorporating binary and multi-class classification on two distinct datasets show that the suggested method performs better than current solutions by producing higher accuracy. It offers high precision and low specificity.

Rani and Shokeen [28] presented a safe ensemble-basis model for fake news identification utilizing blockchain. The identified inadequacies in identifying multi-modal fake news were addressed. A new safe blockchain-dependent deep learning algorithm for identifying fake news was called the Fake News Network (FNNNet). To ensure data integrity as well as learn data depictions, it employs deep learning and blockchain. The ensemble deep learning uses CNN and Bi-LSTM models with CNN capturing the hidden features and Bi-LSTM capturing the sequential data at both directions. Pheme, CrisisLexT6, ISOT databases train the method. It has high recall and low specificity.

Jian et al. [29] developed SA-Bi-LSTM: Self Attention along BiLSTM-Base Intelligent for Exact False News Finding to Ensure Information Integrity at the Social Media. To differentiating between fake and true news, the SA-BiLSTM shows encouraging results. By highlighting important words and phrases, the attention mechanism enables the model to capture crucial characteristics for precise predictions. Cross-validation evaluation yields competitive results for identifying bogus news. The importance of the attention mechanism in recognizing pertinent textual patterns linked to false news was highlighted by ablation research. It attains higher precision and lower recall.

Gorai and Shaw [30] presented semantic variance-dependent feature extraction for false news identification. Cosine similarity, Jaccard distance, and Euclidean distance were used to calculate the vector distances amid the titles and the summarized contents after the article contents have been first summarized using the feature selection. It identifies several features of semantic dissimilarity, leading to a more comprehensive analysis. The findings highlight the value of utilizing semantic distinctions between article names and content, providing a strong substitute for approaches that just concentrate on specific textual elements. It provides high specificity and low f1 score.

Table 1 shows comparison table of literature survey. Although current methods have made a substantial contribution to the identification of fake news, they have drawbacks that compromise detection robustness, contextual comprehension, and generalizability. For example, due to inadequate feature extraction and superficial linguistic modeling, the Tri-Algo guardian ensemble [24] suffered with poor precision despite achieving respectable accuracy. Similar to this, the attribute-wise BiLSTM model [25] did not use multimodal or adversarial training techniques, which is a major reason why it was unable to provide great overall accuracy even if it was able to capture attention-based dependencies. The context-free nature of methods like Word2Vec + LSTM [26] limits their ability to comprehend semantics, whereas text-to-image models [27] added complexity without significantly enhancing specificity. The proposed DEL-BILSTM-AFND model combines CNN for spatial features extraction, Bi-LSTM for sequential dependencies capturing, GANBERT for robust contextual embeddings to overcome these problems. The contextual constraints of Word2Vec-based models are addressed by this hybrid approach, which also improves precision and specificity, two measures that are frequently deficient in earlier research.

Table 1 Comparison table of literature survey

Authors	Approach	Dataset used	Results achieved	Limitations
Vineetha and Nirmala [24]	Tri-Algo Guardian Ensemble	Multiple datasets	Accuracy: 90.19% on Kaggle dataset	Lower precision, lacks contextual embedding
Merryton and Gethsiyal Augusta [25]	Attribute-wise Attention including Bi-LSTM	ISOT Fake news dataset	Precision: 93.02%	Lower accuracy, lacks scalability
Mallik and Kumar [26]	Word2Vec + LSTM	ISOT dataset	F1-score: 90.51%, Accuracy: 91.14%	Context-free model, lacks domain adaptability
Rustam et al. [27]	Text-to-Image Transformation + Custom Models	Binary and multi-class datasets	Accuracy: 88.17%, Precision: 96.32%	Low specificity, computationally intensive
Rani and Shokeen [28]	Secure Blockchain-based Ensemble (FNNet)	PHEME, CrisisLexT6, ISOT dataset	Recall: 71.13%, Precision: 94.13%	Low specificity, high complexity
Jian et al. [29]	SA-Bi-LSTM	ISOT dataset	Precision: 93.16%, F1-score: 95.19%	Lower recall, lacks multimodal integration
Gorai and Shaw [30]	Semantic Difference-based Feature Extraction	ISOT dataset	Specificity: High (93.14%)	Low F1-score, limited to textual similarity

3 Proposed methodology

In this section, DEL-BiLSTM-AFND is discussed. First, input data from the Kaggle and PolitiFact datasets are preprocessed, such as tokenization, in preparation for analysis. The GANBERT model is used to produce high-quality word embeddings to improve textual representation. A CNN extracts significant textual features from the above word embeddings, capturing meaningful patterns. Lastly, a Bi-LSTM network determines whether news articles are authentic or fabricated by examining sequential relationships in the text.

Figure 1 portrays the block drawing of proposed methodology. The diagram portrays a model for detecting fake news that uses an ensemble approach known as EnsembleNet. It begins with news artifacts obtained from the Kaggle and Politi Fact datasets. The refined data are then processed using three deep learning techniques: BiLSTM model for classification, CNN for feature mapping, and GANBERT for word vector generation. To define if the news is reliable or not, the BiLSTM employs learnt patterns. This ensemble strategy utilizes the strengths of various methods to increase misinformation recognition accuracy.

3.1 Problem formulation

Consider a dataset of news stories labeled as O_M . Each news piece is assigned a class designation, with fake news marked 0, and authentic news denoted 1. A news artifact is a

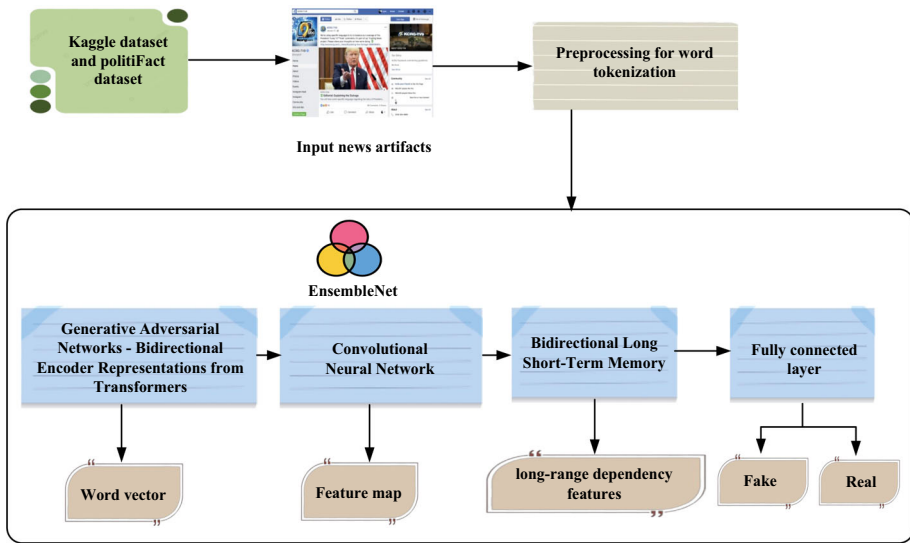


Fig. 1 Proposed DEL-BILSTM-AFND for automated fake news detection

collection of words and sentences that relate to a specific category. Consider the following news artifact $ob_l \in O_M$ comprises j sentences, which are represented as $xe_1, xe_2, xe_3, \dots, xe_o$. Each phrase has the terms designated as given in Eq. (1),

$$X_E = \{xe_1, xe_2, xe_3, \dots, xe_o\} \quad (1)$$

where X_E represents the collection of words that comprise a news article; the problem formulation takes into account both fake and satirical (ironic) news, as humor can occasionally be misconstrued as genuine content when disseminated on online social networks. The fundamental goal of this investigation is to generate a DL technique that can reliably categorize newspapers as either fake or real, as shown in Eq. (2),

$$ob_{current} = \text{Label}(\text{Fake} \vee \text{Real}) \quad (2)$$

where $\vee$ represents the logical operation, ob represents the single news artifact.

3.2 Data acquisition

The data are collected from ISOT Fake news dataset [31], Politi Fact dataset [32].

3.2.1 ISOT fake news dataset

The dataset has truthful or false news. This is taken through actual world sources; truthful articles are acquired through crawling articles via Reuters.com (News website), false news articles are gathered via untrustworthy websites that are labeled by Politifact (fact-verifying association, USA) along with Wikipedia. The dataset comprises diverse kinds of articles on diverse contents. In which, most articles concentrated on political as well as world news topics. This database comprises two CSV files: (i) “True.csv” covers over 12,600 articles

Table 2 Dataset summary with class distribution and data splits

Dataset	Class	Total samples	Training set (70%)	Validation set (15%)	Testing set (15%)
ISOT	Fake News	12,600	8820	1890	1890
	Real News	12,600	8820	1890	1890
	Total	25,200	17,640	3780	3780
PolitiFact	True, Most	24,000 (4,000 per class)	16,800	3600	3600
	True, Half				
	True, Most False, False, Pants Fire				
	Total	24,000	16,800	3600	3600

from reuter.com, (ii) “Fake.csv” covers over 12,600 articles through various fake news outlet resources.

3.2.2 PolitiFact dataset

The PolitiFact Dataset is a gathering of fact-checked statements from the PolitiFact website that are primarily utilized for research on misinformation detection and natural language processing. It includes remarks by politicians, prominent people, and organizations, as well as related truth ratings like true, mostly false, false, true, half true, pants on fire. The dataset often includes metadata such as speaker information, statement context, sources, and dates, which provide useful insights into political debate. Table 2 shows the dataset specification and class distribution.

3.2.3 Preprocessing stage

The effective data preprocessing is required to natural language processing applications such as fake news identification. A comprehensive preparation pipeline is applied to raw news articles from the PolitiFact and ISOT datasets to format, standardize, and sanitize the text data for best compatibility with GANBERT and later deep learning layers. This process included several steps are as follows:

Text cleaning To ensure consistency, all text data are changed to lowercase. Uniform Resource Locators (URLs), Hyper Text Markup Language (HTML) tags, user mentions, and non-ASCII characters were eliminated. Additionally removed are special letters, superfluous whitespace and numerals that had no discernible contribution to semantics.

Stop word removal Natural Language Toolkit (NLTK) stop words list is employed to exclude frequently used stop words that do not significantly improve semantic content. By reducing noise, this step enhances the model’s capacity to focus on more revealing words.

Stemming Porter stemming reduce to their base or root form, which aids in vocabulary reduction. If necessary, word net lemmatization is applied to preserve the words’ linguistic meaning.

Tokenization To comply with GANBERT's standards, the cleaned text was subsequently tokenized into words and sub words using the BERT tokenizer. By doing this, the compatibility of input representations with transformer-based embeddings was guaranteed. This step improves the quality of word embeddings, allowing for more precise feature extraction in the next deep learning stages.

3.3 EnsembleNet model

To process text data efficiently, first it needs to be converted into numerical forms. Word embedding is one such widely used technique for converting words into vector representations. GANBERT is employed in the proposed EnsembleNet model to generate word embeddings. GANBERT not only creates embeddings but also identifies words with similar meaning, so that synonyms are represented uniformly. Rather than concentrating on single words alone, it maintains sentence integrity, which enhances contextual understanding. These word embeddings, generated through the Article Embed Structure, are subsequently passed to a CNN layer, which applies convolutions and filters to extract important features. Activation functions introduce nonlinearity, which enhances feature relationships, while a pooling layer reduces dimensionality. The feature extract structure within the proposed model manages this chain of operations. After CNN feature extraction, a Bi-LSTM layer is employed after the completely linked layer with softmax activation to classify. This last step ensures proper discrimination of real as well as fake news through the integration of the finest of convolutional and sequential processing methods.

3.3.1 Article embed structure using generative adversarial networks with bidirectional encoder representations from transformers model

In this section, GANBERT [33] is discussed. GANBERT is chosen as the embedding module due to its ability to generate semantically rich and context-aware word representations through adversarial training. As GANBERT incorporates adversarial training, which is especially advantageous for unbalanced datasets and low-resource scenarios two major problems in false news detection it is especially selected for this work. GANBERT enhances built-in are then increased through negative training. By gathering complex words and expression associations, the accuracy improves to detect automatic fake news. GANBERT continued DistilBERT in class imbalances and noise -related tasks, and supported the choice. Although models like RoBERTa offer good normal performances, the inherent method is required to handle imbalance in fake news data sets, which usually have little examples of manipulation or misleading information. These built-in are then increased through negative training. By gathering complex words and expression associations, the gambled accuracy improves to detect automatic fake news. GANBERT continued to rota and distilbert in class imbalances and noise related tasks and supported the choice. Although models like RoBERTa offer good normal performances, the inherent method is required to handle imbalance in fake news data sets, which usually have little examples of manipulation or misleading information.

Figure 2 shows GANBERT model [36]. The generator (M) accepts random noise as input and creates synthetic embeddings (E) that resemble real data patterns. Meanwhile, real data (H) are processed using BERT to produce high-quality word embeddings. The model then classifies the embeddings (y) to determine whether they come from actual news or produced samples, hence boosting the network's capacity to detect fake news through representation learning.

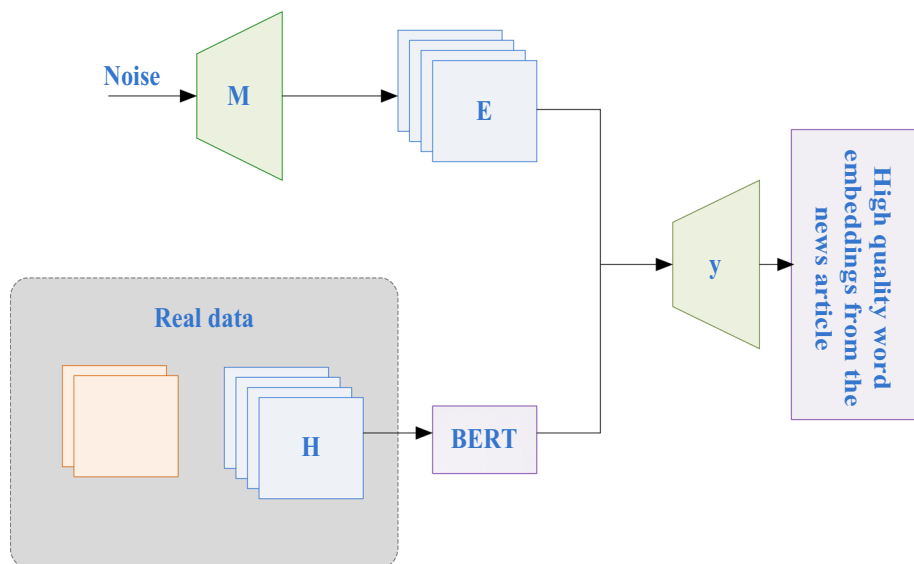


Fig. 2 Architecture of GANBERT model

A true sample and a false sample is given in Eqs. (3) and (4),

$$M_{\text{real}} = \log E(y_{\text{real}}; \theta_e) \quad (3)$$

$$M_{\text{fake}} = \log (1 - E(y_{\text{real}}; \theta_e)) \quad (4)$$

where y_{real} represents the true BERT embeddings of minority-class samples, M_{real} represents the real data, E discriminator function; θ_e represents the generator. The overall discriminator loss is given in Eq. (5),

$$M_E = M_{\text{real}} + M_{\text{real}} \quad (5)$$

where M_E represents the input noise generator; the distribution of the real data is given in Eq. (6),

$$M_H = \log(1 - E(H(a; \theta_h); \theta_e)) \quad (6)$$

where a represents the random noise scalar, θ_h represents the generator function, M_H represents the synthetic data. To guarantee greater detection accuracy, better generalization and enhanced representation learning particularly for difficult and uncommon scenarios in fake news detection GANBERT is chosen. Once GANBERT generates contextual embeddings, these embeddings represent each token in a news article with rich semantic and contextual information.

3.3.2 Feature extracts structure utilizing convolutional neural network

Convolutional Neural Network [34] is applied to extract key textual features by the word embeddings. CNN is chosen because of its ability to extract local patterns and spatial information from word embeddings. CNN layers are quite effective in spotting n-gram or phrase-level

textual cues that could point to misleading language. It is also appropriate for high-throughput, low-latency processing required for real-time detection due to its parallel computation capabilities. CNN's provide considerable benefits in detecting fake news by rapidly evaluating text patterns and structures. Their capacity to capture contextual linkages between words allows them to accurately recognize deceptive information. Furthermore, they improve resilience to hostile manipulations, ensuring dependable findings in a continually changing digital context. The basic purpose of utilizing CNNs in false news identification is to categorize content based on linguistic and contextual clues. Another goal is to increase detection speed and accuracy, making real-time identification possible. Finally, CNNs aim to improve digital literacy by reducing the propagation of disinformation across several internet channels.

Figure 3 exemplifies structure of CNN. It starts with an input layer that transforms text data into an image-like representation or embedding matrix. Convolutional layers use filters to discover patterns in the text, followed by pooling layers that downsample the feature maps, preserving important information while reducing dimension. After being flattened, the retrieved feature maps pass through completely connected layers, allowing for a more in-depth considerate of difficult textual relations. Finally, the output layer uses learnt representations from the feature extraction to define if the news is trustworthy or not.

The convolutional layer is given in Eqs. (7) and (8),

$$A^m = X^m * B^{m-1} + c^m \quad (7)$$

$$B^m = \text{ReLU}(A^m) \quad (8)$$

where A^m implicates output of the convolutional layer, X^m represents filter weights, B^{m-1} embodies preceding layer's input feature map, c^m epitomizes bias term. The max pooling is given in Eq. (9),

$$B^m = \text{MaxPooling}(B^{m-1}, \text{Filter size, stride}) \quad (9)$$

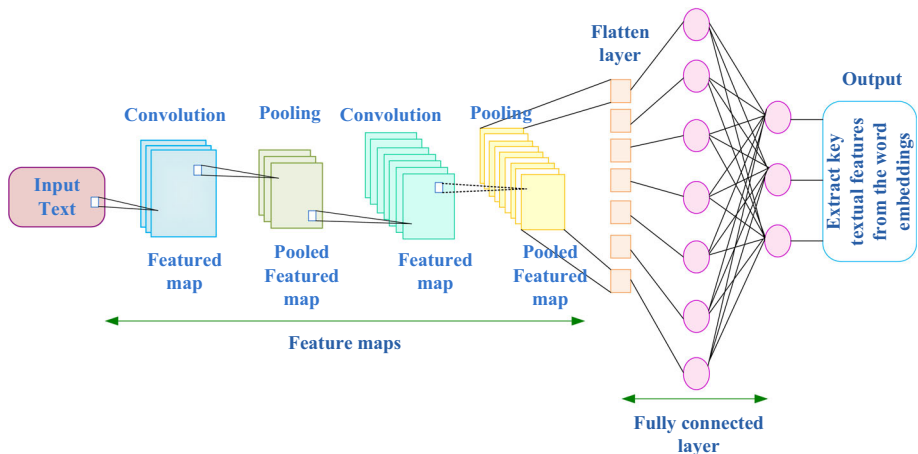


Fig. 3 Architecture of CNN

where B^m represents the input for remainder block, B^{m-1} represents the residual mapping. The dropout operation is given in Eq. (10),

$$A_{\text{Dropout}}^m = \text{Dropout}(B^m, \text{dropout rate}) \quad (10)$$

The softmax function for the output layer is given in Eq. (11),

$$Q(z = j|y) = \frac{f^{Zj}}{\sum_{k=1}^{1000} f^{Zk}} \quad (11)$$

where Q represents the input features, y represents the intermediate feature map. CNN extract key textual features from word embeddings.

3.3.3 News classification using bidirectional long short-term memory model

The Bi-LSTM [35] identify newspapers as truthful or fake with respect to sequential dependencies. BiLSTM enhances the detection of fake news by reading text in each forward as well as backward direction, achieving a better context. This bidirectional approach facilitates better comprehension of language variations and latent patterns in deceptive data. By processing forward and backward sequences, BiLSTM networks are better at capturing long-range dependencies including the contextual flow of text. Contextualizing a remark inside a broader story is often essential when detecting fake news. Bi-LSTM is able to comprehend the structure of the data and determine whether the sentiment development or logical flow indicates dishonest intent through BiLSTM. In addition, BiLSTM manage long-distance relationships very effectively, which suits it for grading complex reports. Consequently, there is a much higher classification rate, enabling much more accurate detection of disinformation. The ultimate purpose of employing BiLSTM in detecting fake news is the enhancement of text analysis via rich contextual understanding. The model incorporates complex linguistic constructs to differentiate true and false data. The other objective is enhancing adaptation so that detection can be done across diverse topics and writing styles. Last but not least, BiLSTM aims to enhance automatic fact-checking systems, thus reducing misinformation spread in online environments.

Figure 4 portrays structure of Bi-LSTM. BiLSTM is an effective neural network model for automated fake news identification. Unlike regular LSTM's, BiLSTM's process text in each backward and forward directions, capturing more contextual meaning. This dual-layer technique helps the model better understand word dependencies, which improve classification accuracy. The LSTM layers extract significant elements from the text, which are then analyzed by a sigmoid activation function to determine whether news is true or fraudulent; by using sequential data. The long-term historical data are given in Eq. (12),

$$g_o = \alpha(X^g \cdot [i_{o-1}, y_o] + c_g) \quad (12)$$

Consider X^g as weight, α as function for activating sigmoid; g_o as forget gate, i_{o-1} as previously hidden state, c_g as prejudice of forget gate, y_o as gates handle input data. The input gate operation is given in Eq. (13),

$$j_o = \alpha(X_j \cdot [i_{o-1}, y_o] + c_j) \quad (13)$$

where c_j represents the new candidate values, j_o represents the input gate, X_j represents the sigmoid function. The current input and hidden state is given in Eq. (14),

$$\overline{D}_o = \tanh(X_f \cdot [i_{o-1}, y_o] + c_d) \quad (14)$$

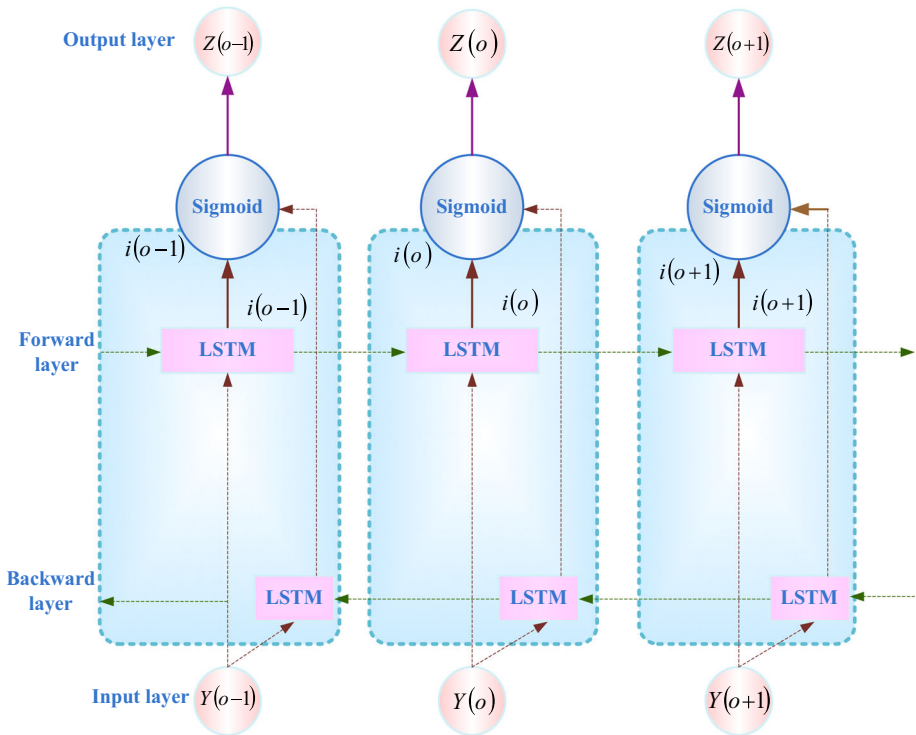


Fig. 4 Architecture of Bi-LSTM

where $\overline{D}_o$ represents the cell state, X_f represents the input sequence, c_d refers output components. The tangent activation process is given in Eq. (15),

$$D_o = (g_o \odot D_{o-1}) + (j_o \odot \overline{D}_o) \quad (15)$$

where D_o represents the cell state, D_{o-1} represents the old state, $\odot$ represents the multiplication of elements. It considers output from the preceding layer and the cell state. It is given in Eq. (16),

$$p_o = \alpha(X_p \cdot [i_{o-1}, y_o] + c_p) \quad (16)$$

where p_o represents the sum of the output components, X_p represents the output gate's weights and bias levels. Bi-LSTM categorizes news articles false or real. Figure 5 shows architecture diagram. Table 3 shows the hyper parameter in CNN and BiLSTM.

4 Results and discussion

The experimental outputs of DEL-BILSTM-AFND method are deliberated here. The DEL-BILSTM-AFND method is implemented in Python language with various inbuilt libraries using pandas to handle data frames, matplotlib for plotting graphical representations, scikit-learn to implement deep algorithms. To access the efficiency of the proposed approach,

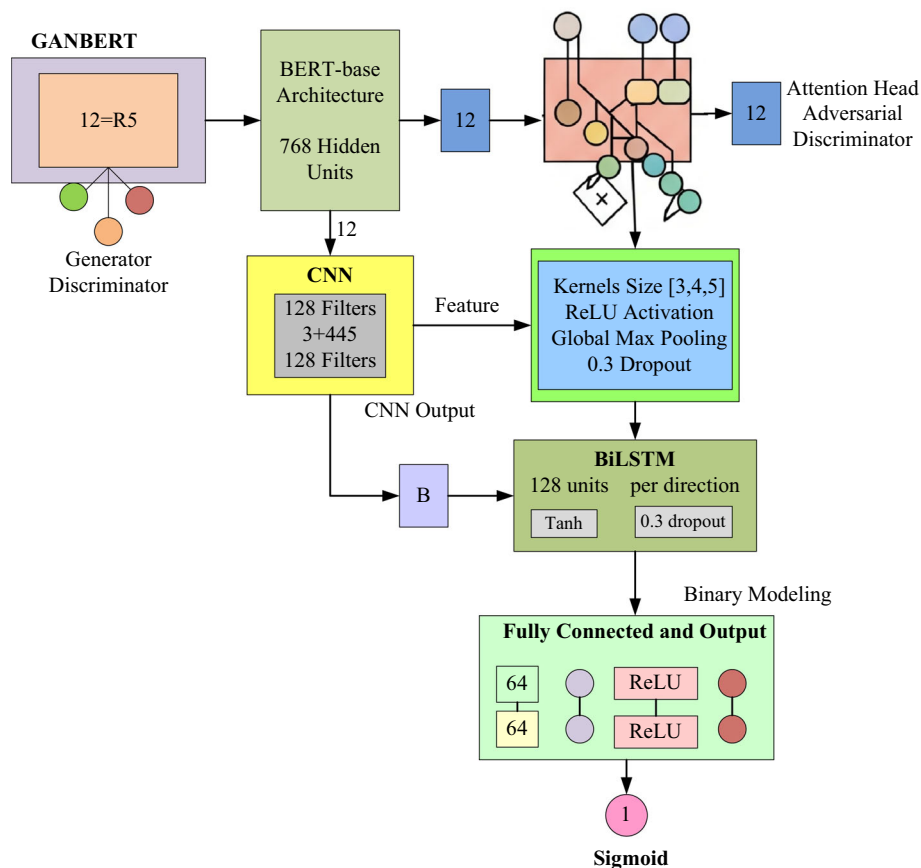


Fig. 5 Architecture diagram for the proposed GANBERT-CNN-BiLSTM ensemble method

Table 3 Summary of hyper parameter modifications in CNN and Bi-LSTM

Component	Hyperparameter	Conventional value	Proposed value
CNN	Number of filters	64	128
CNN	Kernel size	5	3
CNN	Stride	2	1
CNN	Pooling type	Max Pooling	Global max pooling
Bi-LSTM	Number of units	32	64
Bi-LSTM	Activation function	Tanh	ReLU
Bi-LSTM	Dropout rate	0.5	0.3
Bi-LSTM	Bidirectional mode	Unidirectional	Bidirectional

Table 4 Hyper parameter configuration

Component	Hyperparameter	Value
Training	Batch size	100
	Learning rate	0.01
	Optimizer	Mini-batch Gradient Descent
	Loss function	Binary focal cross-entropy
	Epochs	50
Regularization	Dropout rate (CNN/BiLSTM)	0.3
Activation	Hidden layers	ReLU
	Output layer	Sigmoid

accuracy, precision, recall, f1-score, specificity are studied. The results derived from the performance metrics are compared with existing models.

Table 4 depicts hyper parameter configuration. Global max pooling decreases the representation to the topmost features, enhancing the efficiency of computation but lower kernel size (3) ensures sensitivity to short linguistic phrases. The model learns past and future dependencies due to its bidirectional nature, which is vital in understanding causal and logical connections in news articles. A 64-unit design takes into account both over fitting control and the potential to learn. The last dense layer serves as the bridge between outputs and features. The sigmoid activation allows the ability to predict binary probabilities. Random disabling of the neurons using dropout in training prevents over fitting.

4.1 Performance measures

It is necessary for choosing the optimal classifier. The following metrics evaluate the efficacy of the DEL-BILSTM-AFND method.

4.1.1 Accuracy

It assesses how accurate the categorization method's projections are overall. The ratio of accurate projections to total cases is defined by Eq. (17),

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{FN} + \text{TP} + \text{FP} + \text{TN}} \quad (17)$$

TP represents True Positive, TN denotes True Negative, FN signifies False Negative, and *FP* implies False Positive.

4.1.2 Precision

It is a statistic that measures how accurate the classification method's positive predictions are. It describes the ratio of FPs to the sum of TPs using Eq. (18),

$$\text{Precision} = \frac{\text{TP}}{\text{FP} + \text{TP}} \quad (18)$$

4.1.3 Recall

Recall is used in classification tasks to evaluate how well a model recognizes each and every pertinent event in the dataset. This is calculated through Eq. (19),

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{TN}} \quad (19)$$

4.1.4 F1-score

This leads to methods with improved sensitivity, challenges for methods through improved specificity. This is determined by Eq. (20),

$$F1 - \text{score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (20)$$

4.1.5 Specificity

Specificity refers to the methods capability to accurately recognize negative cases among all actual negative cases. This is determined by Eq. (21),

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (21)$$

4.2 Performance analysis of ISOT Fake news Dataset

Table 5 shows simulation results of DEL-BILSTM-AFND technique. Here, the proposed DEL-BILSTM-AFND approach is analyzed to the existing methods, respectively.

Table 5 displays a comparative analysis for fake news ISOT dataset. The comparison analysis demonstrates that the different model architectures and embedding strategies work for detecting fake news. Conventional techniques such as Word2Vec and TF-IDF, exhibit respectable accuracy but frequently have poorer f1- score or recall. Contextual understanding is enhanced by deep learning combinations such as CNN + Bi-LSTM with pre-trained embeddings (e.g., GloVe, BERT), although recall and precision still have to be traded off. The DEL-BILSTM-AFND model combines CNN and Bi-LSTM with GANBERT, performs noticeably better than previous studies on all measures, demonstrating its resilience in managing challenging fake news recognition tasks.

Figure 6 depicts confusion matrix for ISOT fake news dataset. The method's capacity to discriminate fake and real news appears in this binary confusion matrix. With zero false negatives, the program discovered exactly 5497 of all 3477 fake news and 5500 legitimate news articles. The incredibly less false positive frequency is shown from the fact that only three legitimate news pieces are accidentally labeled as a finger. The great efficiency and dependence of the model in differentiation fake and correct news are confirmed by high general accuracy and recall of the matrix accurately, demonstrated its strength and better generalization.

Table 5 Benchmark table for ISOT fake news dataset

Authors	Embedding technique	ML/DL model used	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
Vineetha and Nirmala [22]	TF-IDF	Ensemble (SVM, RF, NB)	90.19	93.02	70.20	68.32
Merryton and GethsiyalAugasta [23]	GloVe	CNN + Attribute-wise Bi-LSTM	85.05	84.18	74.13	75.07
Mallik and Kumar [24]	Word2Vec	LSTM	91.14	70.56	80.53	90.51
Rustam et al. [25]	Text-to-Image Transformation	CNN + Custom Vision Model	88.17	96.32	90.67	73.18
Rani and Shokeen [26]	Random Init + Metadata	CNN + Bi-LSTM + Blockchain Ensemble	70.01	94.13	71.13	65.14
Jian et al. [27]	BERT	Self-Attention + Bi-LSTM	79.16	93.16	79.45	95.19
Gorai and Shaw [28]	Semantic Difference Vectors	Cosine + Jaccard + Euclidean Similarity	85.32	95.13	70.23	94.14
DEL-BiLSTM-AFND (Proposed)	GANBERT (BERT + GAN)	GANBERT + CNN + Bi-LSTM Ensemble	99.40	98.20	98.43	98.25

4.3 Performance analysis of politiFact dataset

Figures 7–11 show the simulation results of DEL-BiLSTM-AFND techniques. Here, the efficiency of the DEL-BiLSTM-AFND is compared with existing TLA-FND [21], WEC-FND [22] and MCCF-FND [23] models, respectively.

Figure 7 shows performance of accuracy. The DEL-BiLSTM-AFND model continuously surpassed the current techniques in all six honesty areas. The proposed model is more reliable and accurate in identifying subtle patterns of bogus news to the successful combination of CNN for local feature extraction, BiLSTM aimed at sequential learning, and GANBERT aimed at rich contextual embeddings. The proposed DEL-BiLSTM-AFND attains 21.34, 19.47 and 23.48% higher accuracy for true; 20.35, 24.62 and 21.56% higher accuracy for mostly-true; 23.16, 21.74 and 21.72% higher accuracy for half-true; 23.67, 21.82 and 18.36% higher accuracy for mostly-false; 18.15, 21.53 and 23.48% higher accuracy for false; 21.63%, 23.78, 21.80% higher accuracy for pants-fire; which is analyzed with existing TLA-FND, WEC-FND and MCCF-FND methods.

Figure 8 shows performance of precision. The DEL-BiLSTM-AFND model minimizes false positives while correctly detecting fake news, resulting in high precision. This is because only very misleading content is identified as phony thanks to the combined strength of CNN

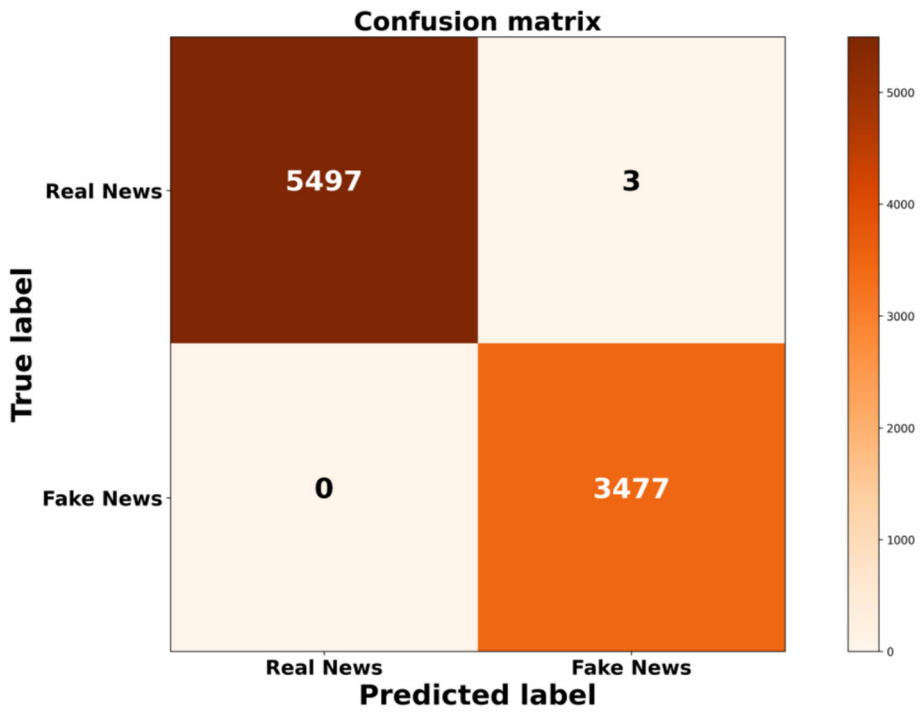


Fig. 6 Confusion matrix for ISOT fake news dataset

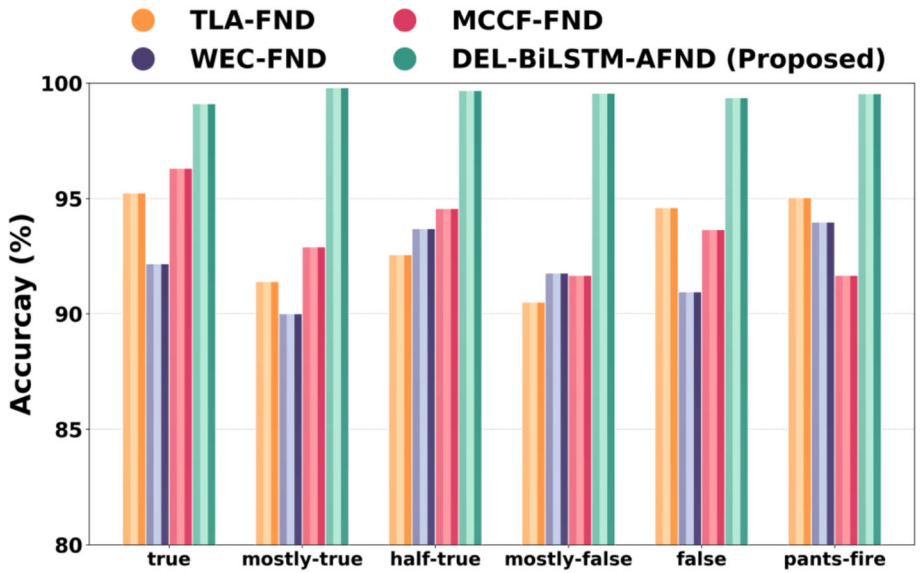


Fig. 7 Performance evaluation of accuracy

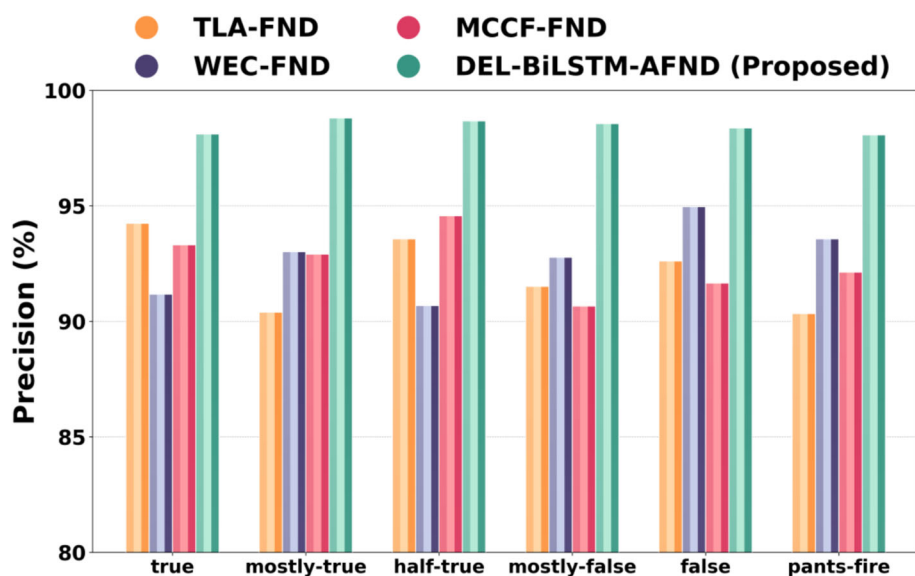


Fig. 8 Performance of precision

feature extraction and GANBERT embeddings. This increases the model's predictability by preventing it from incorrectly classifying legitimate news as fake. The proposed DEL-BiLSTM-AFND attains 19.15, 22.31 and 20.62% higher precision for true; 21.46, 23.68 and 18.44% higher precision for mostly-true; 21.73, 23.84 and 21.62% higher precision for half-true; 20.33, 23.52 and 21.72% higher precision for mostly-false; 19.45, 25.73 and 21.74% higher precision for false; 20.16, 23.46, 21.73% higher precision for pants-fire; which is analyzed with existing TLA-FND, WEC-FND and MCCF-FND methods.

Figure 9 depicts performance of recall. The recall identifies every incident of real false news. By using BiLSTM's sequential analysis, DEL-BiLSTM-AFND exhibits enhanced recall and can identify subtle deceptive patterns across time and structure. This guarantees that extremely few false reports are missed. The proposed DEL-BiLSTM-AFND attains 20.31, 24.64 and 22.58% higher recall for true; 21.57, 22.73 and 19.36% higher recall for mostly-true; 21.47, 24.58 and 19.73% higher recall for half-true; 20.32, 23.85 and 19.64% higher recall for mostly-false; 17.23, 22.63 and 25.29% higher recall for false; 20.52%, 24.47%, 22.68% higher recall for pants-fire; which is analyzed with existing TLA-FND, WEC-FND and MCCF-FND methods.

Figure 10 shows performance of f1 score. The proposed model performs well overall, as evidenced by its strong F1-score values, which strike a compromise amid precision and recall. DEL-BiLSTM-AFND is robust across all false news categories because it combines the contextual strength of GANBERT, the spatial feature mapping of CNN and temporal interdependence for BiLSTM. This results in excellent detection accuracy and low misclassification. The proposed DEL-BiLSTM-AFND attains 23.10, 21.19 and 24.21% higher f1 score for true; 17.24, 19.56, 21.13% better f1 score for mostly-true; 19.45, 20.32% and 22.18% higher f1-score for half-true; 21.67, 18.46, 20.55% better f1-score for mostly-false; 22.83%, 18.38%, 21.29% better f1-score for false; 24.78, 21.47, 20.85% better f1-score for pants-fire analyzed with existing TLA-FND, WEC-FND and MCCF-FND methods.

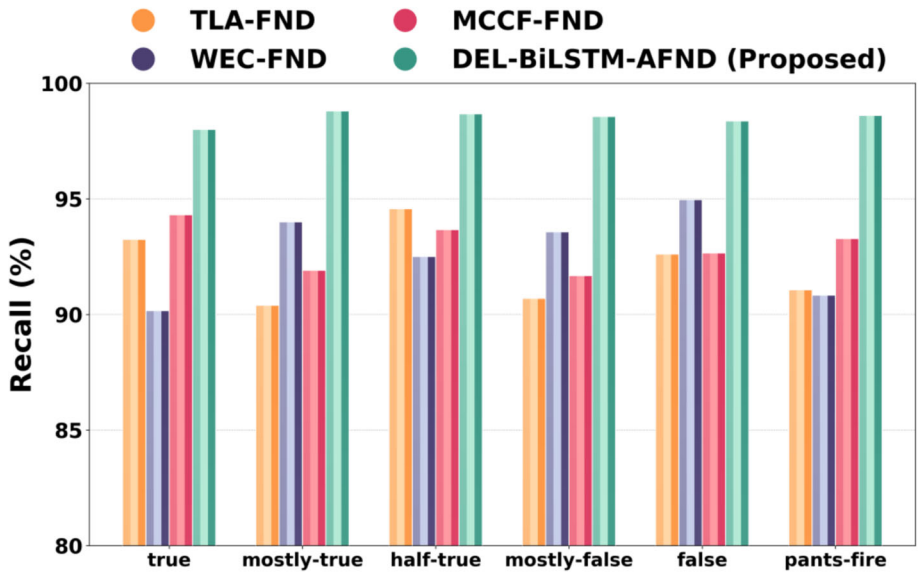


Fig. 9 Performance of recall

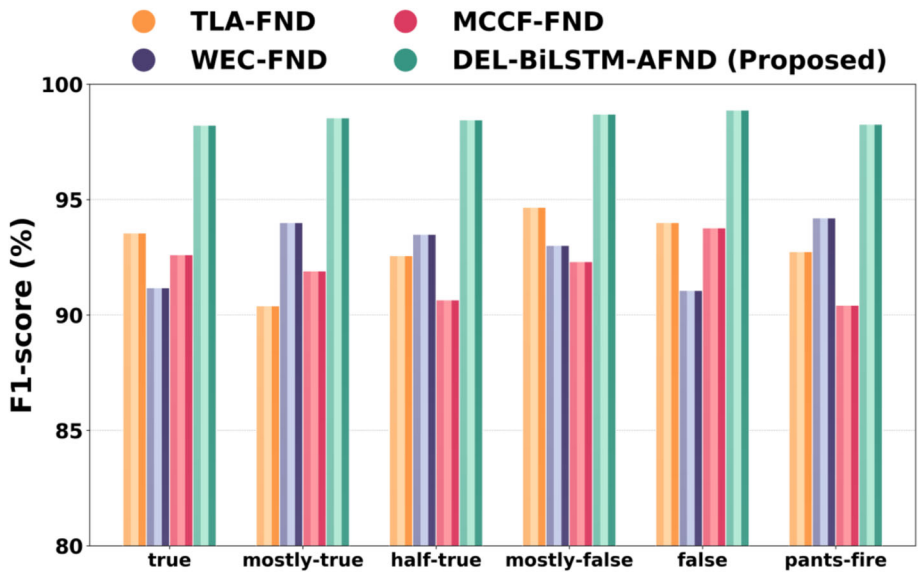


Fig. 10 Performance of F1 score

Figure 11 shows performance of specificity. The model's capacity to accurately identify reliable news and refrain from misclassifying it as fake is reflected in its high specificity. To effectively filter out authentic news information with the least amount of error, DEL-BiLSTM-AFND makes use of contextual knowledge and deep semantic representations. Because of this, it is particularly trustworthy when it comes to maintaining dependable information. The proposed DEL-BiLSTM-AFND attains 21.43, 20.32 and 19.52% higher specificity for true; 18.21, 23.41 and 21.65% higher specificity for mostly-true; 22.36, 20.46 and 23.74% higher specificity for half-true; 20.58, 23.34 and 17.27% higher specificity for mostly-false; 21.73, 23.46 and 20.36% higher specificity for false; 22.51, 24.55, 20.82% higher specificity for pants-fire compared with existing TLA-FND, WEC-FND and MCCF-FND methods.

Table 6 depicts the performance evaluation of error rate. In the proposed ensemble method combining GANBERT and BiLSTM, the error rate compared to individual models was significantly reduced. This indicates the high reliability in differences among actual and false

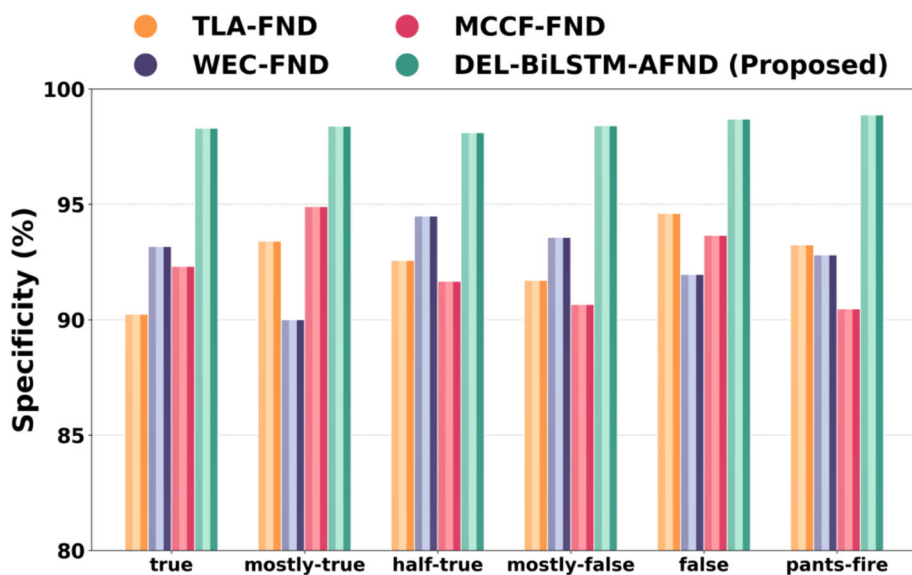


Fig. 11 Performance of specificity

Table 6 Performance evaluation of error rate

Class	TLA-FND (%)	WEC-FND (%)	MCCF-FND (%)	DEL-BiLSTM-AFND (proposed) (%)
True	18.27	20.67	15.52	0.80
Mostly-True	41.21	19.41	26.65	0.75
Half-True	24.36	28.46	19.74	0.66
Mostly-False	27.58	15.34	16.27	0.62
False	30.73	16.46	13.36	0.64
Pants-Fire	20.80	22.75	18.64	0.60

Table 7 Performance evaluation of computational time

Model	Computational time (sec)
TLA-FND	208
WEC-FND	167
MCCF-FND	112
DEL-BILSTM-AFND (Proposed)	99

news. The low error rate shows better generalization and reduces misclassification, which validates the efficiency of the architecture. The proposed DEL-BILSTM-AFND attains 5.27, 8.67 and 7.52% less error rate for true; 7.21, 9.41 and 16.65% lower error rate for mostly-true; 13.36, 8.46 and 12.74% less error rate for half-true; 21.58, 19.34 and 21.27% less error rate for mostly-false; 9.73, 11.46 and 17.36% lower error rate for false; 13.51, 19.55, 20.82% lower error rate for pants-fire; which is analyzed with existing TLA-FND, WEC-FND and MCCF-FND methods.

Table 7 shows the Performance evaluation of Computational time. The proposed DEL-BILSTM-AFND model efficiently integrates GANBERT with BILSTM in a parallel ensemble structure and reduces computational time to 99 s, reducing redundant computation. Unlike traditional sequential pipelines, this design enables rapid processing without compromising accuracy. The model also uses customized data flow and initial convergence techniques, which contribute to better training efficiency on basic methods.

Figure 12 depicts confusion matrix for PolitiFact dataset. The performance of a classification mode in several classes of six true-related labels is shown by this confusion matrix. The model predicted most of the data in each category, as shown by diagonal dominance. The identified successful 5999 of 6,000 and 5500 of 5500 “true” cases 5498. Only a “Pant fire” example was accidentally classified as “true” and two “half -true” examples were expected to be “mostly false.” The misleading was extremely rare. Although some subtle materials overlap each other due to confusion between adjacent sections, the model typically indicates good accuracy and recalls in all categories, indicating reliable performance to distinguish between fine grabs. The factors lack the external knowledge or temporary reference, which limits the model’s ability to validate time-sensitive or fine claims. These rare errors highlight areas where external facts can further improve accuracy with sources, temporary arguments or multimodal signals. Model is affected by keywords or syntactic signals that are misleading in training data.

4.4 Statistical analysis

Table 8 lists ANOVA to estimate the presentation variations among various models by comparing the means of existing models. This analysis scrutinizes variations among both between and within the methods to recognize the reliability of the DEL-BILSTM-AFND effectiveness.

Table 9 depicts the k-fold cross-validation. The fivefold cross-convenience results demonstrate the stability and strength of the proposed DEL-BILSTM-AFND model. The accuracy in all folds was high, from 99.30 to 99.50%, with average 99.40%. Accurate, recall, and F1-score also showed minimal variation, indicating strong generalization capacity. These results confirm that the model performs firmly in various data division to detect fake news.

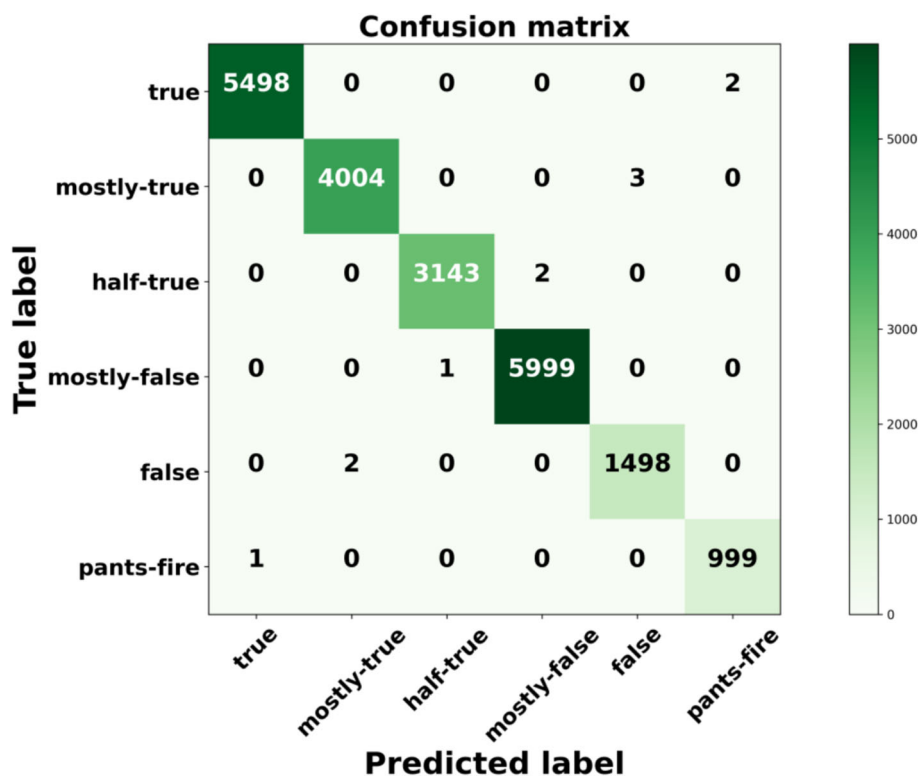


Fig. 12 Confusion matrix for PolitiFact dataset

Table 8 ANOVA examination

Source of variation	Sum of squares	Degrees of freedom	Mean square	F-value	<i>p</i> -value
Between groups	354.76	3	118.25	42.87	< 0.0001
Within 'roups	82.74	36	2.30		
Total	437.50	39			

A *p* value < 0.0001 and an F-value of 42.87 propose that the group means differ statistically significantly. It appears that there is far more variety between the groups than within them, as indicated by the huge mean square value for between groups (118.25) compared to within groups (2.30). This demonstrates that the component under test significantly impacted the result.

4.5 Discussion

The proposed DEL-BILSTM-AFND model effectively integrates various deep learning methodologies to boost detection of false news on online platforms. The model offers various and practical application with the utilization of Kaggle and PolitiFact datasets. Employing GANBERT to achieve high-quality word embeddings strengthens semantic representation of news articles, thus leading to enhanced feature extraction. CNN enhances these embeddings

Table 9 k-fold cross-validation

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Fold 1	99.30	98.10	98.30	98.15
Fold 2	99.45	98.25	98.55	98.30
Fold 3	99.35	98.10	98.40	98.25
Fold 4	99.50	98.30	98.50	98.40
Fold 5	99.40	98.25	98.40	98.35
Average	99.40	98.20	98.43	98.25

by retaining meaningful text patterns, while Bi-LSTM ensures precise classification by analyzing sequential text dependencies. This ensemble approach enhances robustness and is a possible substitute for autonomous FND. To measure the performance the DEL-BILSTM-AFND, the study compared with existing TLA-FND, WEC-FND and MCCF-FND models. The goal of artificial false news detection is to create AI-powered systems that scan text, images, and sources to recognize and mark misinformation with high accuracy. These systems use machine learning, natural language processing also fact-checking databases to minimize the spread of deceptive information and encourage trustworthy journalism. Significant generalization capabilities are provided by the GANBERT embeddings and sequence-aware architecture in the proposed DEL-AFND-DP model. The model is successfully applied to different domains with the correct domain-specific training data, even though it was tested on datasets that comprised broad news and political data. The framework's use of contextual embeddings and sequential modeling allows it to adapt to various linguistic styles, terminologies and deceptive strategies, making it ideal for cross-domain false news detection applications.

4.5.1 Limitations

While the EnsembleNet shows strong performance, it does not completely solve the fake news problem. The model depends more on high quality, domain-specific training data and cannot normalize wrong information well. It also struggles with fine language, satire and cultural variations. Additionally, the dress design increases computational demands, limiting its use in real-time or low-resources settings.

5 Conclusion

The DEL-BILSTM-AFND model was proposed in this paper for automated false news detection on digital platforms. The model successfully caught the linguistic and contextual details of deceptive material by combining CNN for extracting local textual features, Bi-LSTM for modeling sequential dependencies and GANBERT for semantic-rich embeddings. The proposed model achieving 99.4% accuracy, 98.2% precision, 98.43% recall, and 98.25% F1-score on ISOT fake news database. Similarly, DEL-BILSTM-AFND outperformed techniques such as TLA-FND, WEC-FND, and MCCF-FND in the multi-class PolitiFact dataset, achieving up to 24.78% better F1-score and 23.48% greater accuracy across various honesty categories. This work's main contributions are the creation of successful hybrid architecture,

the use of adversarial training to boost classification in the face of data imbalance to improve text clarity. Together, these developments provide a more scalable, effective and generalizable methodology for detecting fake news, which advances the discipline. In practice, the proposed model integrated into public information dashboards, social media content filters and real-time news verification systems to slow the spread of fake news. Its capacity to accurately identify false news promotes better public conversation, safeguards democratic processes and lessens the likelihood of societal disturbance brought on by misleading or manipulative information. The proposed method significantly advances the larger social objective of fostering information integrity in digital communication by increasing automated detection's accuracy and credibility. The model will be expanded in future research to handle multimodal data including images and videos, allowing for more thorough fake news detection. To reduce the misclassification rate in future work incorporating explainable AI (XAI) methods like SHAP or LIME can increase user trust and promote transparency. Also help in identifying which features or tokens contribute to misclassifications. This feedback will be used to retrain the model more effectively on confusing samples.

Author contribution Mrs. Brinda B. M -(Corresponding Author) -Original draft preparation Dr. Rajan C -Supervision Dr. Geetha K 3– Supervision.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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